

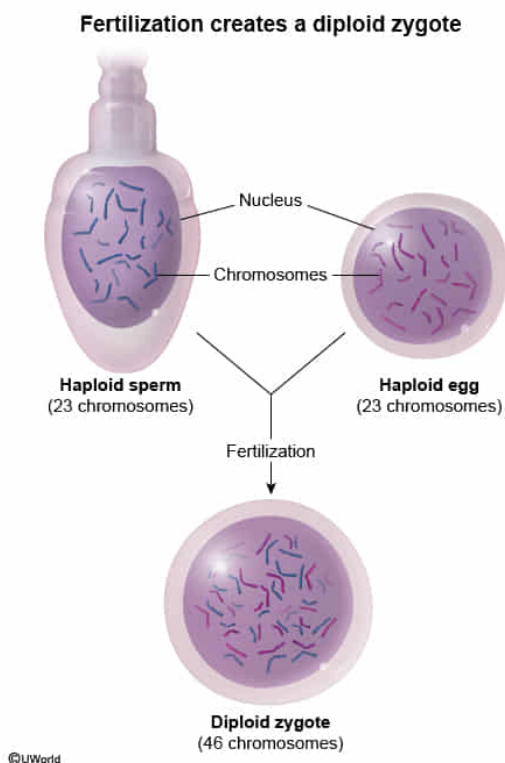
Human sperm and oocytes are substantially different in all of the following aspects EXCEPT:

- ☐ A. cell volume. [4%]
- ☐ B. maturation stage at birth. [8%]
- ✓ ☒ C. the number of nuclear chromosomes contributed to a zygote. [75%]
- ☐ D. the rate at which they are produced. [11%]

Correct

76% answered correctly

Explanation:



Ploidy is defined as the number of homologous **chromosome sets** in the nucleus of a cell. In the human body, with the exception of sperm and egg cells (reproductive cells), nucleated cells are diploid and known as somatic cells. Somatic cells have 46 individual chromosomes that appear as 23 pairs. Each somatic cell has two sets/copies ($2n$) of each chromosome. These include 22 pairs of **homologous** chromosomes (**autosomes**) and 1 pair of sex chromosomes.

In contrast, a mature **egg** and **sperm** are both **haploid** ($1n$), meaning that each contains 22 (unpaired) autosomes and 1 sex chromosome, resulting in 23 chromosomes. During **fertilization**, the egg and sperm form a zygote with a nucleus containing the full complement of 46 chromosomes. These 46 chromosomes are found in pairs, and in each pair one chromosome comes from the mother and one comes from the father. Accordingly, egg and sperm cells are similar in ploidy because they both are haploid cells and **contribute the same number of nuclear chromosomes** to a zygote.

(Choice A) **Sperm** are the smallest cells in the human body and mostly contain a nucleus, an acrosome, mitochondria, and a negligible amount of cytoplasm. In contrast, eggs are the largest human cells and have organelles and a large amount of cytoplasm.

(Choice B) During gestation, the germ cells of the embryo first undergo numerous **mitotic divisions** to proliferate. Next, they are primed for **meiosis**, a type of cell division that results in the formation of gametes. However, the testes of newborn boys contain only immature germ cells (spermatogonia) that have not progressed beyond the mitotic stages. These immature spermatogonia begin to undergo meiosis at puberty (early teen years) when the gonads mature. In contrast, newborn girls have primary oocytes that are arrested at prophase of meiosis I and remain in this state until ovulatory cycles begin at puberty.

(Choice D) In the adult male, millions of sperm cells are produced daily after the onset of puberty. However, females are born with their full complement of oocytes, and only a few mature during each menstrual cycle, which occurs approximately once every 28 days.

Educational objective:

Egg and sperm cells are haploid cells that contribute an equal number of chromosomes to a zygote during fertilization.

Time spent:0 sec

Subject:
Biology

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System:
Reproduction